

A Knowledge Distillation Framework for Edge-Aware Optimization of Intelligent Data

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Abstract—Intelligent systems deployed in safety-critical environments must process large volumes of sensory data under stringent constraints on computation, latency, and energy. These challenges are particularly acute in edge-based applications such as unmanned aerial vehicles operating in search-and-rescue scenarios, where onboard resources are limited and access to large, manually annotated datasets is often impractical. This paper proposes an edge-aware, annotation-free knowledge distillation framework for efficient edge deployment. In contrast to prior work focusing on task-specific distillation or centralized data curation, our framework provides a generalizable, annotation-free pipeline that leverages zero-shot capabilities of foundation models to generate training data and produces a student model validated under strict CPU-only, edge-simulated conditions.

Index Terms—intelligent data, foundation models, knowledge distillation, edge computing, efficient inference.

I. INTRODUCTION

Intelligent systems increasingly rely on large volumes of data to support monitoring, incident detection, and decision-making in safety-critical environments, where timely and reliable information is essential [1]. In such contexts, raw sensor streams must be transformed into *intelligent data*, i.e., semantically enriched representations that can be processed efficiently under operational constraints. This requirement is particularly critical in time-sensitive scenarios, where intelligent data pipelines must balance accuracy, latency, and resource efficiency on resource-constrained platforms, especially at the edge [2].

Within this broader landscape, intelligent perception systems deployed on mobile and autonomous platforms constitute a representative and highly demanding application domain. Such systems operate in dynamic, unstructured, and often safety-critical environments, where rapid situational awareness is essential for reliable decision-making. Vehicles operating at the edge, across aerial, ground, and maritime contexts, enable continuous sensing and wide-area coverage, providing rich visual and multimodal data from environments that are difficult or impractical to monitor centrally [3].

Despite these operational advantages, edge-deployed platforms are inherently constrained by limited onboard computational resources, strict latency requirements, and highly variable operating conditions [4], [5]. These constraints significantly complicate the direct deployment of high-capacity perception models and motivate the need for edge-aware intelligent data processing and optimization strategies that balance semantic richness with predictable and efficient inference.

Beyond computational limitations, the development of effective intelligent data systems is further constrained by the reliance on large-scale, manually annotated datasets [6]. High-quality annotation of complex visual phenomena, such as fire and smoke in long-range or elevated viewpoints, is time-consuming, costly, and difficult to scale, particularly in rapidly evolving safety-critical scenarios. This dependence on manual labeling limits generalization and adaptability, hindering the widespread deployment of intelligent systems across diverse real-world environments [7].

To address these challenges, this work investigates an edge-aware framework for the optimization of intelligent data processing based on knowledge distillation. The main contributions are summarized as follows:

- An *edge-aware optimization framework for intelligent data processing* that explicitly addresses computational, latency, and resource constraints typical of embedded and edge-based systems.
- The use of *semantically enriched representations generated by foundation models as intelligent data*, enabling effective training without reliance on large-scale manually annotated datasets.
- A *knowledge distillation strategy* for transferring semantic knowledge from high-capacity foundation models to lightweight student networks, enabling efficient deployment on resource-constrained platforms, including CPU-only devices.
- A comprehensive evaluation on a challenging fire segmentation task demonstrates that the proposed framework achieves *strong semantic fidelity* (Dice = 0.685), high prediction stability (Consistency IoU = 0.747), and near real-time CPU inference (8.13 FPS), without relying on any manual annotations during training.

Overall, the presented approach illustrates how knowledge distillation can optimize intelligent data pipelines for edge deployment, bridging the gap between high-capacity perception models and real-world constraints. The proposed framework introduces a generalizable, annotation-free workflow using zero-shot foundation models, with explicit evaluation under CPU-only conditions. Although demonstrated on fire segmentation, the pipeline is task-agnostic and applicable to a wide range of perception tasks.

The remainder of this paper is organized as follows. Section II reviews related work on intelligent data processing

under resource constraints. Section III presents the proposed edge-aware optimization framework. Section IV describes the case study and experimental setup. Section V reports the results. Section VI concludes the paper and outlines future work.

II. RELATED WORK

Based on the existing literature reviewed in the following section, research on intelligent data processing can be broadly categorized into three main directions, according to the stage of the intelligent data pipeline they primarily address:

- *Knowledge distillation for efficient learning*, leveraging teacher-student paradigms to improve model efficiency, robustness, and generalization.
- *Annotation-efficient intelligent data generation*, reducing reliance on manual labeling through automated and self-supervised strategies.
- *Edge-aware intelligent data processing*, enabling robust operation under constraints on computation, latency, and energy.

Within this landscape, knowledge distillation is widely used to improve the efficiency and robustness of learning-based models. Prior work explores feature-level and multi-step distillation strategies for object detection and instance segmentation [8], as well as distillation-enhanced monocular depth estimation to improve robustness and generalization in fixed surveillance settings [9]. In more heterogeneous environments, self-distillation has been applied to support cross-domain and incremental learning, such as fire detection across multi-platform sensing modalities [10]. While effective for task-specific performance and robustness, these approaches typically focus on individual models or objectives rather than system-level intelligent data pipelines.

In parallel, several studies emphasize reducing the dependency on large-scale manually annotated datasets by improving the efficiency of intelligent data generation. Semi-automated annotation pipelines combined with active learning enable effective 3D LiDAR semantic segmentation in autonomous railway systems using limited labeled data [11]. Similar challenges are addressed in heritage conservation, where hybrid and multimodal AI approaches are analyzed to handle geometric complexity and heterogeneous point cloud data, highlighting the importance of data acquisition strategies and informed algorithm selection [12]. In industrial-scale mapping applications, intelligent data curation and self-supervised learning further mitigate annotation noise and improve feature extraction accuracy through noise-aware filtering and multi-headed architectures, without extensive manual re-annotation [13]. These approaches primarily target data quality and annotation efficiency, often assuming centralized processing and ample computational resources.

More recent work focuses on intelligent data processing under strict computational constraints in mobile and autonomous platforms. Lightweight detection models with knowledge distillation enable efficient UAV-based fire monitoring [14], while efficiency-oriented perception pipelines in autonomous driving

reduce unnecessary data processing through semantic and BEV representations [15]. Despite their effectiveness, these approaches remain application-specific and do not jointly address learning efficiency, intelligent data generation, and edge deployment.

Most prior work either (a) uses KD but requires manual labels, (b) reduces annotation but assumes cloud-level resources, or (c) is edge-aware but application-specific and label-dependent. None offer a fully annotation-free pipeline using foundation models as teachers for edge-specific distillation. Motivated by these limitations, this work proposes a unified framework that integrates annotation-efficient intelligent data generation with edge-aware knowledge distillation for resource-constrained deployment.

III. PROPOSED FRAMEWORK

This section introduces an edge-aware intelligent data processing framework based on a teacher-student knowledge distillation paradigm. The approach decouples high-capacity semantic reasoning from efficient edge-level inference, enabling deployment under strict computational and latency constraints. The framework follows a two-stage pipeline (Fig. 1). First, a multimodal foundation model acts as a semantic teacher, generating pixel-level segmentation masks from unlabeled images using natural language prompts. These masks serve as intelligent data, eliminating the need for manual annotation. Second, a lightweight student segmentation model is trained on the teacher-generated pseudo-labels and deployed on edge devices using only RGB inputs, with all computationally intensive semantic reasoning performed offline.

A. Knowledge Distillation Method

In the proposed framework, a high-capacity multimodal foundation model is employed as a semantic teacher, while a lightweight convolutional architecture is selected as the student model to meet edge deployment requirements.

a) Teacher Model: Several text-conditioned segmentation foundation models were evaluated as potential teachers in the proposed knowledge distillation pipeline, including CLIPSeg, SAM, Grounded-SAM, SAM2, and SAM3. While CLIPSeg offers lightweight text-based segmentation, its masks are coarse and lack instance separation. The original SAM and its variants relying on spatial prompts or auxiliary detection modules introduce additional supervision requirements and inference complexity, limiting their suitability for fully annotation-free workflows.

SAM3 was selected as the teacher model due to its native support for concept-level text prompting, strong zero-shot generalization, and ability to generate high-quality instance masks without external supervision. These characteristics align closely with the objectives of annotation-free semantic data generation and efficient knowledge transfer to lightweight student models for edge deployment.

b) Student Model: The student model adopts a DeepLabV3 architecture with a MobileNet backbone, a widely

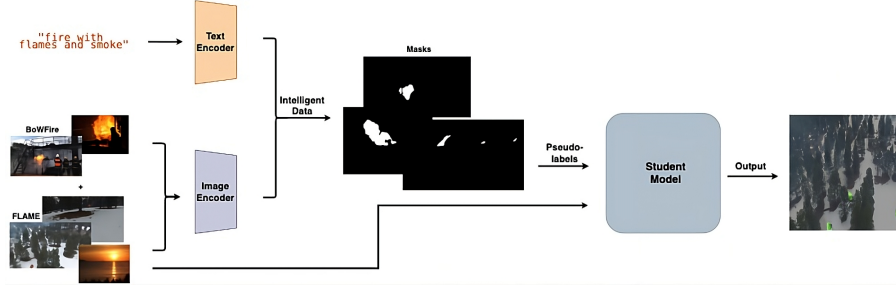


Fig. 1. Overview of the proposed edge-aware teacher-student framework.

TABLE I
COMPARISON OF FOUNDATION MODELS CONSIDERED FOR TEXT-CONDITIONED INTELLIGENT DATA GENERATION.

Model	Prompt Type	Zero-shot	Video	Remarks
CLIPSeg	Text	Yes	No	Lightweight; semantic (non-instance) masks
SAM (original)	Points / Boxes / Masks	Yes	No	Requires adaptation for text-based usage
Grounded-SAM	Text + Boxes / Points	Yes	No	Combines detection with segmentation
SAM2	Multimodal	Yes	Yes	Improved general segmentation; text via adaptations
SAM3 (used)	Concept-level (Text / Image)	Yes	Yes	Native text-conditioned instance segmentation

used and well-documented combination that balances segmentation accuracy and computational efficiency, making it suitable for CPU-only and edge deployment scenarios [16]–[18]. The integration of DeepLabV3 with a MobileNet backbone provides a favorable trade-off between semantic segmentation performance and computational efficiency, making it particularly suitable for CPU-only and edge-oriented deployment scenarios. The student model is trained exclusively using teacher-generated pseudo-labels and operates solely on RGB inputs during both training and inference.

Overall, the proposed framework enables reliable perception under realistic edge constraints by combining intelligent data generation with knowledge distillation-based model compression.

IV. EXPERIMENTAL SETUP

The proposed framework is evaluated through a fire segmentation use case, selected due to its relevance in safety- and security-critical monitoring scenarios. Fire events pose a direct threat to critical infrastructures, urban environments, and natural ecosystems, requiring timely and reliable perception under constrained deployment conditions.

A. Dataset

To assess robustness across heterogeneous environments, two publicly available datasets are combined: BoWFire and FLAME. The BoWFire dataset primarily captures fire incidents in urban and indoor settings, representative of man-made infrastructures such as buildings and industrial facilities. In contrast, FLAME focuses on wildfire scenarios in forest and natural environments, reflecting environmental threats.

The combination of these datasets enables the evaluation of semantic consistency across diverse visual domains. Importantly, no ground-truth segmentation annotations are used during training. All supervision is derived exclusively from

teacher-generated pseudo-labels, demonstrating the applicability of the proposed framework in fully annotation-free and rapidly deployable settings.

B. Metrics

The evaluation protocol employs different metric sets for training, validation, and testing, reflecting both learning effectiveness and deployment robustness.

To assess the quality, consistency, and reliability of the proposed annotation-free distillation framework, a unified set of proxy segmentation metrics is employed. These metrics quantify the spatial extent of predicted regions, the agreement between student predictions and teacher-generated pseudo-labels, and the stability of predictions under simple perturbations.

The spatial coverage of predicted fire regions is measured using the Area Fraction:

$$\text{AreaFrac} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \mathbb{I}(P_{ij} = 1) \quad (1)$$

where P denotes the binary student prediction and $\mathbb{I}(\cdot)$ is the indicator function. This metric captures the proportion of the image classified as fire and provides a normalized measure of predicted spatial extent.

Agreement between student predictions and teacher-generated pseudo-labels is evaluated using the Teacher–Student Intersection-over-Union (IoU) where T denotes the teacher-generated pseudo-label and P the student prediction:

$$\text{IoU}_{TS} = \frac{|P \cap T|}{|P \cup T|} \quad (2)$$

which measures the relative overlap between student and teacher masks, penalizing both false positives and false negatives with respect to the teacher.

In addition, the Dice coefficient is used as a complementary overlap metric:

$$\text{Dice}_{TS} = \frac{2|P \cap T|}{|P| + |T|} \quad (3)$$

Dice is less sensitive to small object sizes and is particularly informative under class imbalance.

Moreover, prediction stability is assessed via Consistency IoU under horizontal flipping:

$$\text{IoU}_{flip} = \frac{|P \cap P^{flip}|}{|P \cup P^{flip}|} \quad (4)$$

where P^{flip} denotes the prediction obtained from a horizontally flipped input and mapped back to the original image orientation. This metric evaluates the robustness and self-consistency of the model under simple geometric perturbations.

Finally, during training, convergence and stability are monitored using the training loss:

$$\mathcal{L}_{train} = \text{CE}(P, T), \quad (5)$$

where $\text{CE}(\cdot)$ denotes the pixel-wise cross-entropy loss computed between the student prediction P and the teacher-generated pseudo-label T , with ignored pixels excluded. This loss quantifies how effectively the student learns to replicate the semantic structure provided by the teacher.

Together, these metrics provide a compact yet comprehensive proxy evaluation of semantic fidelity, robustness, and consistency in the absence of ground-truth annotations.

a) Inference Performance Metrics: In addition to segmentation quality, inference efficiency is evaluated to assess the suitability of the distilled student model for edge deployment. All performance measurements are conducted under batch size one, which reflects realistic real-time edge perception scenarios.

The following inference metrics are reported:

- **Inference Latency (ms/frame):** average wall-clock time required to process a single image, measured during test-time execution.
- **Throughput (FPS):** frames per second achieved during continuous inference, computed as the inverse of average latency.

These metrics provide an upper bound on deployment cost under constrained execution environments and are particularly relevant for security- and safety-critical monitoring systems, where predictable latency and continuous operation are required.

V. RESULTS

The evaluation follows a unified, lifecycle-oriented analysis of the proposed framework, examining how intelligent data are generated, transferred, and deployed under realistic edge constraints. All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 3060 GPU with 12 GB of memory, using a fixed and deterministic dataset split (70/10/20) to ensure reproducibility.

The analysis first assesses candidate foundation models as semantic generators, focusing on pseudo-label reliability and consistency. It then evaluates knowledge transfer to a lightweight student model via distillation, and finally examines the edge feasibility of the distilled model through inference efficiency under constrained execution conditions.

A. Teacher Model Selection and Consistency Analysis

Table I summarizes the candidate foundation models considered for text-conditioned intelligent data generation. CLIPSeg is used as a lightweight semantic baseline, while SAM3 represents a higher-capacity model with native concept-level text conditioning and instance-level segmentation.

Structural reliability is assessed using IoU_{flip} between original and horizontally flipped predictions, serving as a proxy for pseudo-label stability under mild perturbations.

As shown in Table II, SAM3 achieves markedly higher flip-consistency ($\text{IoU}_{flip} = 0.77$) than CLIPSeg (0.46), indicating more stable and coherent pseudo-label generation under simple input perturbations. This difference reflects SAM3's stronger semantic alignment and instance-level segmentation capability, whereas CLIPSeg produces coarser and less stable masks.

TABLE II
MEAN IoU_{flip} AND AREA FRACTION ACROSS CANDIDATE TEACHER MODELS

Model	IoU_{flip}	Area Fraction
CLIPSeg	0.46	0.008
SAM3	0.77	0.009

The Area Fraction remains low for both models (0.008–0.009), reflecting the spatial sparsity of fire regions in the evaluated datasets due to elevated viewpoints and long-range sensing. Consequently, Area Fraction is treated as a descriptive property of the data rather than a performance metric.

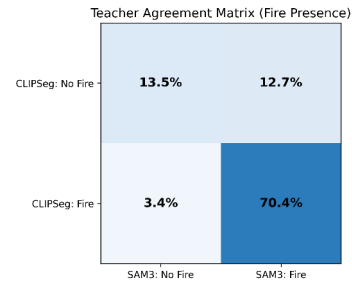


Fig. 2. Image-level agreement matrix between CLIPSeg and SAM3 for binary fire / no-fire decisions.

Figure 2 presents the image-level agreement between the two models. Overall agreement reaches 83.9%, with SAM3 detecting fire in an additional 12.7% of images where CLIPSeg predicts no fire, while the opposite occurs in only 3.4% of cases. This asymmetric disagreement indicates broader semantic coverage and higher sensitivity of SAM3 to fire-related visual patterns.

These results motivate the selection of SAM3 as the teacher model for knowledge distillation, as it combines higher pseudo-label stability with broader semantic coverage compared to the baseline.

B. Knowledge Distillation and Student Learning

Following teacher selection, SAM3 is employed to generate pseudo-labels for training a lightweight DeepLabV3-based student model, enabling the evaluation of semantic knowledge transfer to a resource-efficient architecture.

Figure 3 shows the evolution of the training loss during distillation. The loss decreases rapidly in early epochs and stabilizes around epoch 39, despite training being scheduled for 60 epochs, indicating convergence and limited benefit from further optimization.

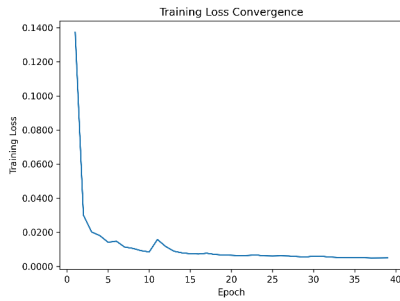


Fig. 3. Training loss convergence of the student model during knowledge distillation.

Quantitative evaluation using teacher-student agreement metrics confirms effective knowledge transfer. At convergence, the student achieves a validation IoU of 0.568 and a Dice score of 0.673 (Table III), demonstrating strong alignment with the teacher’s semantic outputs despite reduced model complexity.

TABLE III

VALIDATION PERFORMANCE OF THE STUDENT MODEL AT CONVERGENCE.

Metric	Value
Validation IoU	0.568
Validation Dice	0.673

Figure 4 provides a comparison between the original RGB input, the teacher-generated pseudo-label, and the student prediction overlay. Despite the small spatial extent of fire regions and the absence of manual annotations, the student accurately reproduces the teacher’s semantic decisions, indicating effective transfer of high-level information.



Fig. 4. Qualitative comparison of RGB input, teacher pseudo-label (SAM3), and student prediction overlay.

Overall, the results indicate stable convergence and effective annotation-free knowledge distillation, supporting reliable semantic transfer to a lightweight model suitable for edge deployment.

C. Inference Performance and Edge Feasibility

An image-level comparison between teacher pseudo-labels and student predictions is illustrated in Fig. 5 through a binary fire / no-fire confusion matrix. The results indicate a very high level of agreement between the two models (**99.2%**). Specifically, 84.9% of the images are jointly identified as containing fire, while 15.0% are consistently classified as fire-free. Importantly, no false negatives are observed, meaning that the student never fails to detect fire when the teacher predicts its presence. Only **0.8%** of the samples correspond to cases where the student predicts fire while the teacher does not, a behavior commonly reported in the knowledge distillation literature and often attributed to the student surpassing the teacher under noisy supervision.

Teacher-Student Confusion Matrix (Percentage)

Teacher: No Fire	14.9%	0.8%
Teacher: Fire	0.0%	84.3%
	Student: No Fire	Student: Fire

Fig. 5. Image-level confusion matrix illustrating agreement between teacher pseudo-labels and student predictions for fire presence.

TABLE IV

TEACHER-STUDENT AGREEMENT AND CONSISTENCY METRICS.

Metric	Mean Value
Teacher-Student IoU	0.581
Teacher-Student Dice	0.685
Flip Consistency IoU	0.747

As summarized in Table IV, the student model achieves substantial agreement with the teacher-generated pseudo-labels, attaining a mean Teacher–Student IoU of 0.581 and a Dice score of 0.685. These values indicate that the student effectively reproduces the spatial structure of the teacher’s semantic predictions despite its reduced capacity. In addition, the higher Flip Consistency IoU of 0.747 demonstrates that the student’s predictions remain stable under simple geometric perturbations, reflecting robust and self-consistent behavior. The combination of image-level agreement and pixel-level metrics highlights a clear distinction between semantic correctness and boundary precision. Although IoU and Dice scores are not maximal, the student consistently detects all fire instances at the image level, with deviations primarily

TABLE V
INFERENCE PERFORMANCE UNDER CPU-ONLY EDGE-ORIENTED
EXECUTION.

Metric	Value
Inference Latency (ms/frame)	123.03
Throughput (FPS)	8.13

attributed to boundary smoothing and noise suppression. High flip-consistency further confirms the robustness of the learned representation.

To approximate realistic edge deployment conditions, inference was evaluated under CPU-only execution with batch size one and a single processing thread. Under this conservative setting, the student model achieves an average latency of 123.03 ms per frame, corresponding to 8.13 FPS (Table V) demonstrating predictable near real-time performance suitable for continuous, event-driven on-device monitoring tasks. Since no hardware acceleration is used, these results represent a lower bound on performance, with higher throughput expected on dedicated edge AI platforms.

Overall, the results confirm that the distilled student model combines strong semantic agreement, high prediction stability, and predictable low-latency inference, supporting its suitability for edge-based intelligent data processing.

VI. CONCLUSIONS AND FUTURE WORK

The deployment of intelligent systems in safety-critical, resource-constrained environments requires data-efficient, edge-aware perception pipelines. While foundation models offer strong semantic understanding, their computational demands limit direct edge deployment. This work addresses this gap by transferring semantic knowledge under realistic edge constraints.

An edge-aware knowledge distillation framework is introduced, enabling annotation-free training of lightweight models using foundation-model-generated pseudo-labels. Results on fire segmentation show stable convergence, strong teacher-student agreement, and near real-time CPU performance. Although demonstrated on fire segmentation, the framework is designed as a general methodology applicable across diverse perception tasks.

Evaluation uses teacher-student agreement as a proxy aligned with the annotation-free objective, while not reflecting ground-truth accuracy, SAM3 provides a reliable semantic reference.

Future work will extend the proposed framework beyond fire segmentation to broader perception tasks, including multi-class scene understanding and anomaly detection. Model compression techniques, such as pruning and quantization, will be explored to reduce latency, memory footprint, and energy consumption for efficient edge deployment. Evaluation on embedded platforms (e.g., NVIDIA Jetson) and real-world scenarios is planned to assess scalability and operational readiness, along with comparative analysis using additional lightweight segmentation models and alternative student configurations.

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